Basin MCMC Assignment 1

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The purpose of this assignment is to break down the development of the initial basin probability distribution and probability in the Basin MCMC algorithm.

Throughout this assignment, we use the following notation:

$$\mathcal{X} := \bigcup_{n=0}^{\infty} \mathcal{X}_n,$$

where

$$\mathcal{X}_n := \left([0, X_{\max}] \times (0, D_{\max}] \times [0, \infty) \times [-1, 1]\right)^n.$$

Question 1. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_1 : \mathcal{X}_1 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. a good approximation to the log posterior in the region of high probability around θ_0;
3. easy to compute defining quantities from θ_0;
4. easy to sample from;
5. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. We present the solution as follows:

1. Define f_1 as the second-order Taylor polynomial of the log posterior about θ_0;
2. Derive a formula for the gradient $g = \nabla \log p(\theta_0 \mid y)$;
3. Derive a formula for the negative Hessian $P = -\nabla^2 \log p(\theta_0 \mid y)$;
4. State the final formula for f_1.

1. For $\theta \in \mathcal{X}_1$, the log posterior is given by

$$\log p(\theta \mid y) = -\frac{1}{2\sigma^2} \|y - A(\theta)\|_2^2 + \log p(\theta) + C,$$

where C is some constant independent of θ, and $A : \mathcal{X} \to [0, 1]^{N_t N_x}$ is the forward map. Performing a second-order Taylor expansion about θ_0 (since $\log p(\theta \mid y) \in C^3(U)$ for some open neighbourhood $U \subset \mathcal{X}_1$ of θ_0, provided $\theta_0 \in \mathrm{int}(\mathcal{X}_1)$) gives

$$\log p(\theta \mid y) = \log p(\theta_0 \mid y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0) + O(\|\theta - \theta_0\|^3), \tag{1}$$

where

$$g := \nabla \log p(\theta_0 \mid y) \quad \text{and} \quad P := -\nabla^2 \log p(\theta_0 \mid y).$$

Motivated by (1), we define our approximation to the log posterior $f_1 : \mathcal{X}_1 \to \mathbb{R}$ by

$$f_1(\theta) := \log p(\theta_0 \mid y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0).$$

2. We have that

$$g = \nabla \log p(\theta_0 \mid y) = \nabla (\log p(y \mid \theta_0) + \log p(\theta_0) - \log p(y)) = \nabla \log p(y \mid \theta_0) + \nabla \log p(\theta_0). \tag{2}$$

For $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(y \mid \theta) = \nabla \left(-\frac{1}{2\sigma^2} \|y - A(\theta)\|_2^2 \right) = \frac{1}{\sigma^2} J_A(\theta)^\top (y - A(\theta)),$$

where $J_A(\theta)$ is the Jacobian of the forward map A (restricted to the domain $\mathcal{X}_1$). Recalling that $y = A(\theta_0)$,

$$\nabla \log p(y \mid \theta_0) = \frac{1}{\sigma^2} J_A(\theta_0)^\top (y - A(\theta_0)) = 0. \tag{3}$$

Finally, for $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(\theta) = \nabla \log \left(\lambda_R \mathrm{e}^{-\lambda_r R} \right) = \nabla (-\lambda_R R) = (0, 0, -\lambda_R, 0)^\top. \tag{4}$$

Combining (2), (3), and (4), we have that

$$g = (0, 0, -\lambda_R, 0)^\top.$$

3. The negative Hessian of the log posterior is given by

$$P = -\nabla^2 \log p(\theta_0 \mid y) = -\nabla^2 \log p(y \mid \theta_0) - \nabla^2 \log p(\theta_0). \tag{5}$$

Recalling (4), we have that

$$\nabla^2 \log p(\theta_0) = 0. \tag{6}$$

Next, for $\theta \in \mathcal{X}_1$, one can show that

$$\nabla^2 \log p(y \mid \theta) = -\frac{1}{\sigma^2} J_A(\theta)^\top J_A(\theta) + \frac{1}{\sigma^2} \sum_k (y - A(\theta))_k \nabla^2 A_k(\theta).$$

Again recalling that $y = A(\theta_0)$, we obtain

$$\nabla^2 \log p(y \mid \theta_0) = -\frac{1}{\sigma^2} J_A(\theta_0)^\top J_A(\theta_0). \tag{7}$$

Combining (5), (6), and (7), we have that

$$P = \frac{1}{\sigma^2} J_A(\theta_0)^\top J_A(\theta_0). \tag{8}$$

To obtain a formula for P, we will obtain a formula for $J_A(\theta_0)$. To this end, recall that the $tN_x + x$-th entry of the forward map $A : \mathcal{X} \to [0,1]^{N_t N_x}$ with parameters $\theta = (x_0, d, R, a)^\top \in \mathcal{X}_1$ is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}}e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x-x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2)e^{-(\pi f_0 \tau)^2}.$$

Applying the product and chain rules, we obtain

$$\begin{aligned}
\frac{\partial A(t,x;\theta)}{\partial x_0} &= aB(x)\left[w(t-T(x))\frac{\partial \log B(x)}{\partial x_0} + w'(t-T(x))\frac{-\partial T(x)}{\partial x_0}\right], \\
\frac{\partial A(t,x;\theta)}{\partial d} &= aB(x)\left[w(t-T(x))\frac{\partial \log B(x)}{\partial (d+R)} + w'(t-T(x))\frac{-\partial T(x)}{\partial d}\right], \\
\frac{\partial A(t,x;\theta)}{\partial R} &= aB(x)\left[w(t-T(x))\frac{\partial \log B(x)}{\partial (d+R)} + w'(t-T(x))\frac{-\partial T(x)}{\partial R}\right], \\
\frac{\partial A(t,x;\theta)}{\partial a} &= B(x)\, w(t-T(x)).
\end{aligned} \tag{9}$$

The needed derivatives are given by

$$\begin{aligned}
w'(\tau) &= 2(\pi f_0)^2 \tau (2(\pi f_0 \tau)^2 - 3)e^{-(\pi f_0 \tau)^2}, \\
\frac{\partial \log B(x)}{\partial x_0} &= (x - x_0)\left(\frac{1}{2s(x)^2} + \frac{2\alpha}{s(x)}\right), \\
\frac{\partial \log B(x)}{\partial (d+R)} &= \frac{1}{2(d+R)} - \frac{(d+R)}{2s(x)^2} - \frac{2\alpha(d+R)}{s(x)}, \\
-\frac{\partial T(x)}{\partial x_0} &= \frac{2(x-x_0)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial d} &= -\frac{2(d+R)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial R} &= \frac{2}{v}\left(1 - \frac{(d+R)}{s(x)}\right).
\end{aligned}$$

Vectorising each of the derivatives in (9) gives the four columns of $J_A(\theta_0)$.

4. Recall that for $\theta \in \mathcal{X}_1$, our approximation to the log posterior is given by

$$f_1(\theta) = \log p(\theta_0 \mid y) + g^{\mathsf{T}}(\theta - \theta_0) - \frac{1}{2}(\theta - \theta_0)^{\mathsf{T}} P (\theta - \theta_0).$$

Assuming that P is positive definite and so P^{-1} exists (which holds if and only if $J_A(\theta_0)$ has full rank), we may complete the square to obtain

$$\begin{aligned}
f_1(\theta) &= \log p(\theta_0 \mid y) + g^{\mathsf{T}}(\theta - \theta_0) - \frac{1}{2}(\theta - \theta_0)^{\mathsf{T}} P (\theta - \theta_0) \\
&= \log p(\theta_0 \mid y) - \frac{1}{2}\left[(\theta - \theta_0)^\top P(\theta - \theta_0) - 2g^\top(\theta - \theta_0)\right] \\
&= \log p(\theta_0 \mid y) - \frac{1}{2}\left[(\theta - \theta_0)^\top P(\theta - \theta_0) - 2g^\top(\theta - \theta_0) + g^\top P^{-1} g - g^\top P^{-1} g\right] \\
&= \log p(\theta_0 \mid y) - \frac{1}{2}\left[(\theta - \theta_0 - P^{-1}g)^\top P(\theta - \theta_0 - P^{-1}g) - g^\top P^{-1} g\right] \\
&= \log p(\theta_0 \mid y) + \frac{1}{2}g^\top P^{-1} g - \frac{1}{2}(\theta - \theta_0 - P^{-1}g)^\top P(\theta - \theta_0 - P^{-1}g)
\end{aligned}$$

$$= C' - \frac{1}{2}(\theta - \mu)^\top P(\theta - \mu),$$

where C' is some constant independent of θ, and $\mu := \theta_0 + P^{-1}g$.

Thus, our approximation $f_1(\theta)$ defines a truncated Gaussian on $\mathcal{X}_1$, with mean μ and covariance P^{-1}. In the implementation, the Moore–Penrose pseudoinverse P^+ is used in place of P^{-1} for numerical robustness.

Question 2. [pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\intercal \in \mathcal{X}_1$. Provide a function that approximates the log likelihood $\log p(y \mid \theta)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $\tilde{f}_2 : \mathcal{X}_2 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $\tilde{f}_2(\theta_1, \theta_2) = \tilde{f}_2(\theta_2, \theta_1)$;
3. a good approximation to the log likelihood in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. We present the solution as follows: TODO: Change the presentation once the solution is complete.

1. Present the log likelihood and the forward map;
2. Define a reparameterization;
3. Rewrite the forward map in terms of the reparameterization;
4. Taylor approximate the forward map around the center of the reparameterization;
5. Taylor approximate the forward map around the representative atom;
6. Approximate the Hessian of the forward map around the center of the reparameterization by the Hessian of the forward map around the representative atom;
7. Combine the Taylor approximations to obtain the final formula for $\tilde{f}_2$.

1. For $(\theta_1, \theta_2) \in \mathcal{X}_2$, the log likelihood is given by

$$\log p(y \mid \theta_1, \theta_2) = -\frac{1}{2\sigma^2} \|A(\theta_0) - A(\theta_1) - A(\theta_2)\|_2^2 + C$$

where C is some constant independent of (θ_1, θ_2), and $A : \mathcal{X} \to [0,1]^{N_t N_x}$ is the forward map. Recall that the $tN_x + x$-th entry of the forward map $A : \mathcal{X} \to [0,1]^{N_t N_x}$ with parameters $\theta = (x_0, d, R, a)^\intercal \in \mathcal{X}_1$ is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}} e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2) e^{-(\pi f_0 \tau)^2}.$$

2. Define $g_0 := (x_{00}, d_0, R_0)^\mathsf{T}$, so that we may write the representative atom as $\theta_0 = (g_0, a_0)^\mathsf{T}$. Now, with two atoms $\theta_1 = (g_1, a_1)^\mathsf{T}, \theta_2 = (g_2, a_2)^\mathsf{T} \in \mathcal{X}_1$, where $g_1 := (x_1, d_1, R_1)^\mathsf{T}$ and $g_2 := (x_2, d_2, R_2)^\mathsf{T}$, and $a_1 + a_2 \neq 0$, we will use the following reparameterization:

   - $a := a_1 + a_2$ is the total amplitude,
   - $\lambda := \frac{a_1}{a_1+a_2}$ is the amplitude split,
   - $g := \lambda g_1 + (1-\lambda) g_2$ is the amplitude-weighted center,
   - $h := g_1 - g_2$ is the difference.

   We may recover the original parameters as follows:

   - $a_1 = \lambda a$,
   - $a_2 = (1-\lambda)a$,
   - $g_1 = g + (1-\lambda)h$,
   - $g_2 = g - \lambda h$.

3. Define the function $F : [0, X_{\max}] \times (0, D_{\max}] \times [0, \infty) \to [0,1]^{N_t N_x}$ by

   $$F(g) := A(g, 1),$$

   where $A(g,1)$ is the $N_t N_x$-dimensional vector with $A(t, x; g, 1)$ as the $tN_x + x$-th entry. Notice that $A(g,a) = aF(g)$ for $a \in [-1,1]$. Hence,

   $$\begin{aligned} A(\theta_1) + A(\theta_2) &= a_1 F(g_1) + a_2 F(g_2) \\ &= a\left[\lambda F(g_1) + (1-\lambda) F(g_2)\right] \\ &= a\left[\lambda F(g+\delta_1) + (1-\lambda) F(g+\delta_2)\right], \end{aligned} \tag{10}$$

   where $\delta_1 := g_1 - g = (1-\lambda)h$ and $\delta_2 := g_2 - g = -\lambda h$.

4. Now, for $g \in \operatorname{int}(\mathcal{X}_1)$, (since F is smooth in an open neighbourhood of $g \in \operatorname{int}(\mathcal{X}_1)$) performing a second-order Taylor expansion of F around g gives

   $$F(g+\delta) = F(g) + J_F(g)\delta + \frac{1}{2} H_F(g)[\delta, \delta] + O(\|\delta\|^3) \tag{11}$$

   as $\delta \to 0$, where $J_F(g)$ is the Jacobian of F at g,

   $$J_F(g) = \left[\frac{\partial F(g)}{\partial x_0}, \frac{\partial F(g)}{\partial d}, \frac{\partial F(g)}{\partial R}\right] = \begin{bmatrix} \frac{\partial F_1(g)}{\partial x_0} & \frac{\partial F_1(g)}{\partial d} & \frac{\partial F_1(g)}{\partial R} \\ \vdots & \vdots & \vdots \\ \frac{\partial F_{N_t N_x}(g)}{\partial x_0} & \frac{\partial F_{N_t N_x}(g)}{\partial d} & \frac{\partial F_{N_t N_x}(g)}{\partial R} \end{bmatrix},$$

   and $H_F(g)[\delta, \delta]$ is the $N_t N_x$-dimensional vector with k-th entry

   $$\left(H_F(g)[\delta,\delta]\right)_k = \delta^\mathsf{T} \nabla^2 F_k(g) \delta,$$

   where $F_k(g)$ is the k-th entry of $F(g)$.

   Using the Taylor approximation (11) for each term in (10) gives

   $$\begin{aligned} \lambda F(g+\delta_1) + (1-\lambda) F(g+\delta_2) = F(g) &+ J_F(g)\left(\lambda\delta_1 + (1-\lambda)\delta_2\right) + \frac{1}{2}\left(\lambda H_F(g)[\delta_1,\delta_1] + (1-\lambda) H_F(g)[\delta_2,\delta_2]\right) \\ &+ O(\|\delta_1\|^3 + \|\delta_2\|^3), \end{aligned} \tag{12}$$

   as $(\delta_1, \delta_2) \to 0$. Now, notice that since $\lambda\delta_1 + (1-\lambda)\delta_2 = 0$, we have that

   $$J_F(g)\left(\lambda\delta_1 + (1-\lambda)\delta_2\right) = 0. \tag{13}$$

Next, notice that since $H_F(g)[\delta,\delta]$ is bilinear, we have that

$$\begin{aligned}\lambda H_F(g)[\delta_1,\delta_1]+(1-\lambda)H_F(g)[\delta_2,\delta_2] &= \lambda(1-\lambda)^2 H_F(g)[h,h]+(1-\lambda)\lambda^2 H_F(g)[h,h]\\ &= \lambda(1-\lambda)H_F(g)[h,h]. \end{aligned} \tag{14}$$

Combining (12), (13), and (14) gives

$$\lambda F(g+\delta_1)+(1-\lambda)F(g+\delta_2) = F(g)+\frac{1}{2}\lambda(1-\lambda)H_F(g)[h,h]+O(\|\delta_1\|^3+\|\delta_2\|^3). \tag{15}$$

Since

$$\|\delta_1\|^3+\|\delta_2\|^3 = \left(|\lambda|^3+|1-\lambda|^3\right)\|h\|^3,$$

if we restrict to the region where $|a_1+a_2|>a_{\min}$ for some $a_{\min}>0$ (which should be reasonable otherwise they cancel each other out in the image), so that $|\lambda|, |1-\lambda|<1/a_{\min}$, then we have that

$$O(\|\delta_1\|^3+\|\delta_2\|^3) = O\left(\|h\|^3\right), \tag{16}$$

as $h\to 0$. Overall, combining (15) and (16) gives

$$\lambda F(g+\delta_1)+(1-\lambda)F(g+\delta_2) = F(g)+\frac{1}{2}\lambda(1-\lambda)H_F(g)[h,h]+O\left(\|h\|^3\right) \tag{17}$$

as $h\to 0$ (where the $O\left(\|h\|^3\right)$ remainder is uniform in λ in the region where $|a_1+a_2|>a_{\min}$).

5. Now, for $g_0\in \operatorname{int}(\mathcal{X}_1)$, (since F is smooth in an open neighbourhood of $g_0\in\operatorname{int}(\mathcal{X}_1)$) performing a first-order Taylor expansion of F around g_0 gives

$$F(g_0+\delta) = F(g_0)+J_F(g_0)\delta+O(\|\delta\|^2) \tag{18}$$

as $\delta\to 0$, where $J_F(g_0)$ is the Jacobian of F at g_0, given by

$$J_F(g_0) = \left[\frac{\partial F(g_0)}{\partial x_0}, \frac{\partial F(g_0)}{\partial d}, \frac{\partial F(g_0)}{\partial R}\right] = \begin{bmatrix} \frac{\partial F_1(g_0)}{\partial x_0} & \frac{\partial F_1(g_0)}{\partial d} & \frac{\partial F_1(g_0)}{\partial R}\\ \vdots & \vdots & \vdots \\ \frac{\partial F_{N_tN_x}(g_0)}{\partial x_0} & \frac{\partial F_{N_tN_x}(g_0)}{\partial d} & \frac{\partial F_{N_tN_x}(g_0)}{\partial R}\end{bmatrix}.$$

Applying the approximation (18) to $F(g)$ in (17) gives

$$\begin{aligned}\lambda F(g+\delta_1)+(1-\lambda)F(g+\delta_2) &= F(g_0+\delta_0)+\frac{1}{2}\lambda(1-\lambda)H_F(g)[h,h]+O\left(\|h\|^3\right)\\ &= F(g_0)+J_F(g_0)q+\frac{1}{2}\lambda(1-\lambda)H_F(g)[h,h]+O\left(\|\delta_0\|^2+\|h\|^3\right)\end{aligned} \tag{19}$$

as $(h,\delta_0)\to 0$, where $\delta_0 := g-g_0$ (where the remainder is uniform in λ in the region where $|a_1+a_2|>a_{\min}$).

6. Since we would like an approximation that only depends on the representative atom (g_0,a_0), we will approximate $H_F(g)[h,h]$. (Since F has bounded third derivatives in an open neighbourhood of $g_0\in\operatorname{int}(\mathcal{X}_1)$), we have that

$$H_F(g)[h,h] = H_F(g_0)[h,h]+O(\|\delta_0\|\|h\|^2)$$

as $(h,\delta_0)\to 0$. Combining this with (19) gives

$$\begin{aligned}\lambda F(g+\delta_1)+(1-\lambda)F(g+\delta_2) &= F(g_0)+J_F(g_0)\delta_0+\frac{1}{2}\lambda(1-\lambda)H_F(g_0)[h,h]\\ &\quad +O\left(\|\delta_0\|^2+\|\delta_0\|\|h\|^2+\|h\|^3\right),\end{aligned} \tag{20}$$

as $(h,\delta_0)\to 0$ (where the remainder is uniform in λ in the region where $|a_1+a_2|>a_{\min}$).

7. Finally, with $\eta_0 := a - a_0$, combining (20) and (10) we have

$$\begin{aligned}
A(\theta_1) + A(\theta_2) &= a\big[F(g_0) + J_F(g_0)\delta_0 + \frac{1}{2}\lambda(1-\lambda)H_F(g_0)[h,h] \\
&+ O\left(\|\delta_0\|^2 + \|\delta_0\|\|h\|^2 + \|h\|^3\right)\big] \\
&= (a_0 + \eta_0)\big[F(g_0) + J_F(g_0)\delta_0 + \frac{1}{2}\lambda(1-\lambda)H_F(g_0)[h,h] \\
&+ O\left(\|\delta_0\|^2 + \|\delta_0\|\|h\|^2 + \|h\|^3\right)\big] \\
&= a_0F(g_0) + \eta_0F(g_0) + aJ_F(g_0)\delta_0 + \frac{a}{2}\lambda(1-\lambda)H_F(g_0)[h,h] \\
&+ O\left((1+|\eta_0|)\left(\|\delta_0\|^2 + \|\delta_0\|\|h\|^2 + \|h\|^3\right)\right) \\
&= A(\theta_0) + \eta_0F(g_0) + aJ_F(g_0)\delta_0 + \frac{a}{2}\lambda(1-\lambda)H_F(g_0)[h,h] \\
&+ O\left(\|\delta_0\|^2 + \|\delta_0\|\|h\|^2 + \|h\|^3\right) \qquad (21)
\end{aligned}$$

as $(h, \delta_0, \eta_0) \to 0$ (where the remainder is uniform in λ in the region where $|a_1 + a_2| > a_{\min}$). That is,

$$\begin{aligned}
A(\theta_1) + A(\theta_2) - A(\theta_0) &= (a - a_0)\,F(g_0) + aJ_F(g_0)\,(g - g_0) + \frac{a}{2}\lambda(1-\lambda)H_F(g_0)[h,h] \\
&+ O\left(\|g - g_0\|^2 + \|g - g_0\|\|h\|^2 + \|h\|^3\right) \qquad (22)
\end{aligned}$$

as $(h, g, a) \to (0, g_0, a_0)$ (where the remainder is uniform in λ in the region where $|a_1 + a_2| > a_{\min}$). In fact, for $a_0 \neq 0$, we need not impose separately the restriction that $|a_1+a_2| > a_{\min}$, since $a_1+a_2 = a \to a_0$ implies that for a sufficiently close to a_0, we have $|a_1 + a_2| > |a_0|/2 =: a_{\min}$.

Question 3. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_2 : \mathcal{X}_2 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $f_2(\theta_1, \theta_2) = f_2(\theta_2, \theta_1)$;
3. a good approximation to the log posterior in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant. This should build on the function found in Question 2.

Solution.

Question 4. **[pts.]**

Fix $n \geq 2$. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_n$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_n : \mathcal{X}_n \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable;
3. a good approximation to the log posterior in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant, and this constant need not be common to all n. This should be a generalization of the function found in Question 3.

Solution.

Question 5. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Rewrite the function found in Question 1 in the same form as the functions found in Question 4.

If it is not possible to write the function found in Question 1 in the same form as the functions found in Question 4, explain why. Then, provide an alternative function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$ that is in the same form as the functions found in Question 4.

Solution.

Question 6. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide an approximation of the sequence of probabilities $\{p_n\}_{n\in\mathbb{N}}$ where

$$p_n := P(\theta \in X_n \mid y) = \int_{\mathcal{X}_n} p(\theta \mid y)\mathrm{d}\theta,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0.

You may find it useful to use the functions found in Questions 4 and 5. Note that you need only approximate the sequence up to a common multiplicative constant; that is, it suffices to approximate quantities proportional to $\{p_n\}_{n\in\mathbb{N}}$.

Solution.

Question 7. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide an approximation of the probability p where

$$p :=,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0.

Solution.